

浙江财经大学

课后作业

项 目 名 称	Practice of database design
所 属 课 程 名 称	Database Theory and Application
项 目 类 型	Comprehensive Exercise
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学 院	
班 级	
学 号	
姓 名	
指导教师	Deng Shengchun

Practice of database design

1. E-R modeling and derive relations from E-R diagram

Subject:

There are four entities in one hospital ward management system as following.

- ① Department: Dno(Department No.), Dname(Department Name), Location, Phone.
- ② Ward: Wno (Ward No.), Location.
- ③ Doctor: Eno (Employee No.), Ename(Employee Name), Title(Professional Title), Gender, Birthday
- ④ Patient: Pno(Patient No.), Pname(Patient Name), Gender, Birthday

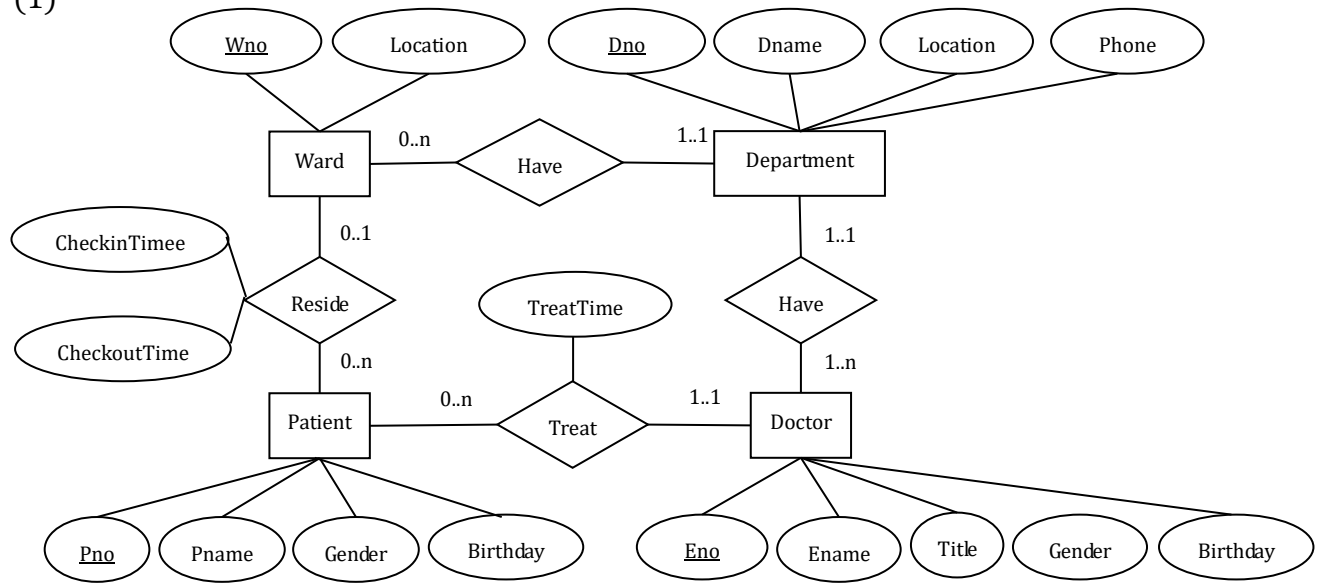
In the above hospital ward management system, you should consider the following business rules.

- ① One department may have many wards and many doctors, but one doctor must always belong to one department, and one ward must belong to one department at a certain time.
- ② One doctor could be responsible for the diagnosis and treatment of many patients, but one patient only has one doctor in charge.
- ③ One ward may have many patients, but one patient can only live in one ward at a certain time.

Requirement:

- (1) Please draw the E-R diagram of the above hospital ward management system.
- (2) Please transform the E-R model (conceptual model) to relational model (logical model), and denote the primary key for each relation.

(1)



(2)

2.1

Department(Dno, Dname, Location, Phone)

primary key: (Dno),

alternate key: (Dname)

foreign key: None.

2.2

Ward(Wno, Location, Dno)

primary key: (Dno, Wno)

alternate key: None

foreign key: (Dno).

2.3

Doctor(Eno, Ename, Title, Gender, Birthday, Dno)

primary key: (Eno)

alternate key: None

foreign key: (Dno).

2.4

Patient(Pno, Pname, Gender, Birthday)

primary key: (Pno)

alternate key: None

foreign key: None.

2.5

Treat(Eno, Pno, TreatTime)

primary key: (Eno, Pno)

alternate key: None

foreign key: (Eno), (Pno).

2.6

Residence(Pno, Wno, CheckinTime, CheckoutTime)

primary key: (Pno, Wno)

alternate key: None

foreign key: (Pno), (Wno).

2. Normalization

Subject:

Consider the following relations that represent the information of an invoice of a product sale:

Invoice_data (I_number, I_date, C_number, C_name, C_city, Item_number, Item_name, Item_qty, Item_price, Total_price).

Each invoice(发票) has a number(I_number), date(I_date), a single customer number(C_number), total price(Total_price) of all items.

Each invoice may list one or more items that the customer bought, and each item in the invoice has a unique item number(Item_number), name of item(Item_name), quantity(Item_qty) and price(Item_price).

Items of product may be listed in more than one invoice, that is to say may be bought by more than one customer.

Each customer has a customer number (C_number), customer name (C_name), customer city(C_city).

Requirement:

- (1) Identify the primary key of above table;
- (2) Identify the functional dependencies represented by the description in the above table;
- (3) Describe and illustrate the process of normalizing the relation shown in the above table to BCNF.

Answer:

(1) The primary key is (I_number, Item_number)

(2) Functional dependencies:

FD1. (I_number, Item_number) $\rightarrow$ I_date [partial functional dependency]

FD2. (I_number, Item_number) $\rightarrow$ C_number [partial functional dependency]

FD3. (I_number, Item_number) $\rightarrow$ C_name [partial functional dependency]

FD4. (I_number, Item_number) $\rightarrow$ C_city [partial functional dependency]

FD5. (I_number, Item_number) $\rightarrow$ Item_name [partial functional dependency]

FD6. (I_number, Item_number) $\rightarrow$ Item_qty [full functional dependency]

FD7. (I_number, Item_number) $\rightarrow$ Item_price [partial functional dependency]

FD8. (I_number, Item_number) $\rightarrow$ Total_price [partial functional dependency]

FD9. I_number $\rightarrow$ I_date [full functional dependency]

FD10. I_number $\rightarrow$ C_number [full functional dependency]

FD11. I_number $\rightarrow$ C_name [full functional dependency]

FD12. I_number $\rightarrow$ C_city [full functional dependency]

FD13. I_number $\rightarrow$ Total_price [full functional dependency]

FD14. Item_number $\rightarrow$ Item_name [full functional dependency]

FD15. Item_number $\rightarrow$ Item_qty [full functional dependency]

FD16. Item_number $\rightarrow$ Item_price [full functional dependency]

(3)

(3.1) the above relation satisfies 1NF, because there is no sub-attribute.

(3.2) because there are some non-primary key attributes are partially functional dependent of the primary key, so it is not satisfies 2NF, we need to decompose the above relation into three relations as following.

① Invoice_detail (I_number, Item_number, Item_qty)

② Invoice (I_number, I_date, C_number, C_name, C_city, Total_price)

③ Item (Item_number, Item_name, Item_price)

In the above three relations, there is non-primary key attribute is partially functional dependent on the primary key, so they all satisfy 2NF.

(3.3) For relation ① Invoice_detail (I_number, Item_number, Item_qty), there is only one FD: (I_number, Item_number) $\rightarrow$ Item_qty, there is no transitive FD, so relation ① satisfy 3NF. At the same time, (I_number, Item_number) is candidate key in FD1, so relation ① satisfy BCNF.

For relation ③ Item (Item_number, Item_name, Item_price), there are two

FDs: (Item_number) $\rightarrow$ Item_name, (Item_number) $\rightarrow$ Item_price. there is no transitive FD, so relation ③ satisfy 3NF. At the same time, (Item_number) is candidate key in above FDs, so relation ③ satisfy BCNF.

For relation ② Invoice (I_number, I_date, C_number, C_name, C_city, Total_price), there are following FDs:

FD1: I_number $\rightarrow$ I_date [full functional dependency]

FD2: I_number $\rightarrow$ C_number [full functional dependency]

FD3: I_number $\rightarrow$ Total_price [full functional dependency]

FD4: C_number $\rightarrow$ C_name [full functional dependency]

FD5: C_number $\rightarrow$ C_city [full functional dependency]

FD6: I_number $\rightarrow$ C_name [transitive functional dependency]

FD7: I_number $\rightarrow$ C_city [transitive functional dependency]

There are two transitive FDs, so relation ② does not satisfy 3NF, we need to decompose the above relation into two relations as following.

④ Invoice (I_number, I_date, C_number, Total_price)

⑤ Customer (C_number, C_name, C_city)

For relation ④ Invoice (I_number, I_date, C_number, Total_price), there are three FDs: (I_number) $\rightarrow$ I_date, (I_number) $\rightarrow$ C_number, (I_number) $\rightarrow$ Total_price. There is no transitive FDs, so relation ④ satisfy 3NF. At the same time, (I_number) is candidate key in above FDs, so relation ④ satisfy BCNF.

For relation ⑤ Customer (C_number, C_name, C_city), there are two FDs: (C_number) $\rightarrow$ C_name, (C_number) $\rightarrow$ C_city. There is no transitive FDs, so relation ⑤ satisfy 3NF. At the same time, (C_number) is candidate key in above FDs, so relation ⑤ satisfy BCNF.

(3.4) Finally, we decompose the original relation into four relations as following;

Invoice (I_number, I_date, C_number, Total_price)

Invoice_detail (I_number, Item_number, Item_qty)

Item (Item_number, Item_name, Item_price)

Customer (C_number, C_name, C_city)